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(@jriva)

Revising Aggregations - The Count Function

The screenshot shows the HackerRank interface for the 'Revising Aggregations - The Count Function' challenge. The problem statement asks for a count of cities in the CITY table with a population larger than 100,000. The input format section provides the schema for the CITY table:

Field	Type
ID	NUMBER
NAME	VARCHAR2(17)
COUNTRYCODE	VARCHAR2(3)
DISTRICT	VARCHAR2(20)
POPULATION	NUMBER

The SQL editor on the right contains the following code:

```
1 SELECT COUNT(*) FROM CITY
2 WHERE POPULATION > 100000;
3
4
```

At the bottom right, the status 'Line: 1 Col: 17' is displayed. Buttons for 'Run Code' and 'Submit Code' are visible.

Revising Aggregations - The Sum Function

The screenshot shows the HackerRank interface for the 'Revising Aggregations - The Sum Function' challenge. The problem statement asks for the total population of all cities in the CITY table where the district is California. The input format section provides the schema for the CITY table:

Field	Type
ID	NUMBER
NAME	VARCHAR2(17)
COUNTRYCODE	VARCHAR2(3)
DISTRICT	VARCHAR2(20)
POPULATION	NUMBER

The SQL editor on the right contains the following code:

```
1 SELECT SUM(POPULATION) FROM CITY
2 WHERE DISTRICT = 'California';
3
```

At the bottom right, the status 'Line: 1 Col: 24' is displayed. Buttons for 'Run Code' and 'Submit Code' are visible.

Revising Aggregations - Averages

Query the average population of all cities in **CITY** where District is **California**.

Input Format

The **CITY** table is described as follows:

Field	Type
ID	NUMBER
NAME	VARCHAR2(17)
COUNTRYCODE	VARCHAR2(3)
DISTRICT	VARCHAR2(20)
POPULATION	NUMBER

SQL Editor:

```
1 SELECT AVG(POPULATION) FROM CITY
2 WHERE DISTRICT = 'California';
```

Line: 1 Col: 24

Buttons: Upload Code as File, Test against custom input, Run Code, Submit Code

The Blunder

Samantha was tasked with calculating the average monthly salaries for all employees in the **EMPLOYEES** table, but did not realize her keyboard's 0 key was broken until after completing the calculation. She wants your help finding the difference between her miscalculation (using salaries with any zeros removed), and the actual average salary.

Write a query calculating the amount of error (i.e., *actual* - *miscalculated* average monthly salaries), and round it up to the next integer.

Input Format

The **EMPLOYEES** table is described as follows:

Column	Type
ID	Integer
Name	String
Salary	Integer

Note: Salary is per month.

Constraints

SQL Editor:

```
1 SELECT CEILING(
2     AVG(DECIMAL(SALARY, 10, 2)) -
3     AVG(
4         DECIMAL(
5             NULLIF(REPLACE(VARCHAR(SALARY), '0', ''), ''),
6             10, 2
7         )
8     )
9 )
10 FROM EMPLOYEES;
11
```

Line: 11 Col: 1

Buttons: Upload Code as File, Test against custom input, Run Code, Submit Code

Top Earners

We define an employee's total earnings to be their monthly *salary* $\times$ *months* worked, and the maximum total earnings to be the maximum total earnings for any employee in the **Employee** table. Write a query to find the maximum total earnings for all employees as well as the total number of employees who have maximum total earnings. Then print these values as 2 space-separated integers.

Input Format

The **Employee** table containing employee data for a company is described as follows:

Column	Type
employee_id	Integer
name	String
months	Integer
salary	Integer

```
1 SELECT
2   MAX(salary * months) AS max_earnings, COUNT(*)
3 FROM Employee
4 WHERE (months * salary) = (SELECT MAX(months * salary)
5 FROM Employee
6 );
7
```

Line: 7 Col: 1

Buttons: Upload Code as File, Test against custom input, Run Code, Submit Code

Solved Assigned Problems

Weather Observation Station 5
Easy, SQL (Intermediate), Max Score: 30, Success Rate: 94.79% Solved

Revising Aggregations - The Count Function
Easy, SQL (Basic), Max Score: 10, Success Rate: 99.47% Solved

Revising Aggregations - The Sum Function
Easy, SQL (Basic), Max Score: 10, Success Rate: 99.60% Solved

Revising Aggregations - Averages
Easy, SQL (Basic), Max Score: 10, Success Rate: 99.78% Solved

The Blunder
Easy, SQL (Basic), Max Score: 15, Success Rate: 97.07% Solved

Top Earners
Easy, SQL (Basic), Max Score: 20, Success Rate: 98.29% Solved

STATUS
☒ Solved
☐ Unsolved

SKILLS
☐ SQL (Basic)
☐ SQL (Intermediate)
☐ SQL (Advanced)

DIFFICULTY
☐ Easy
☐ Medium
☐ Hard

SUBDOMAINS
☐ Basic Select
☐ Advanced Select
☐ Aggregation
☐ Basic Join
☐ Advanced Join
☐ Alternative Queries

<https://www.hackerrank.com/challenges/revising-aggregations-sum/problem?isFullScreen=true>